

KARTIKEYA SINGH

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Professional Summary

AI Software Engineer with production experience building **LLM-powered pipelines**, **Multi-Agent Systems**, and **RAG** architectures at scale. Shipped an event-driven call intelligence system at IndiaMART using **Kafka**, **Gemini 2.5 Flash**, and **LiteLLM** with structured output schemas to eliminate hallucinations. Proficient in end-to-end AI development: retrieval (**LlamaIndex**, **LangChain**, **Qdrant**), orchestration (**LangGraph**), inference optimization (**NVIDIA Triton**, **vLLM**), and evaluation (**Ragas**). Seeking AI/LLM engineering roles focused on building reliable, production-grade generative AI systems.

Education

National Institute of Technology, Patna

B.Tech in Computer Science & Engineering

2021 – 2025

8.83/10 CGPA

Work Experience

Infoedge (Naukri)

Apr 2026 – Present

Senior AI Engineer (NVIDIA Triton, vLLM, Python)

Noida

- Spearheaded the deployment of high-performance ML models utilizing **NVIDIA Triton Inference Server**, optimizing serving infrastructure for production-scale generative AI workloads.
- Implemented dynamic batching and concurrent model execution configurations, drastically cutting down inference **latency** while multiplying request **throughput** to handle high user traffic.
- Architected hardware-aware inference pipelines that significantly reduced overall **CPU load** by offloading bottlenecks to GPUs, maximizing memory bandwidth and compute utilization.

Indiamart Intermesh Ltd

Sep 2025 – Apr 2026

Software Engineer (Golang, Kafka, LangChain, LiteLLM, Ragas)

Noida

- Engineered an event-driven **call intelligence pipeline** using **Kafka**, **Debezium**, and **Cloud SQL (GCP)** to centralize real-time data extraction across thousands of daily seller-buyer calls, deprecating 5+ redundant isolated API processes.
- Integrated **Gemini 2.5 Flash** via **LiteLLM** with Structured JSON Schemas to enforce deterministic model outputs—eliminating hallucinations and enabling accurate extraction of **10+ product attributes** per call transcript.
- Established **Ragas**-based evaluation framework to continuously monitor RAG pipeline quality, tracking faithfulness and answer relevancy metrics across model iterations to guide prompt and retrieval improvements.

TATA 1MG

Jan 2025 – Sep 2025

Backend Development Intern (Python, FastAPI, SQS, MongoDB)

Chandigarh

- Architected a scalable, high-throughput microservice ecosystem using **FastAPI** and **Sanic** with **async I/O**, enabling modular, independently deployable API endpoints under sustained load.
- Integrated **Spike API**, replacing the existing Terra API to stream wearable health data—reducing third-party vendor cost by **\$4,200 annually** while improving data freshness.
- Designed low-latency data APIs with **Pydantic** schema validation and **OAuth 2.0** authentication to deliver fast, reliable user health insights.
- Optimized critical API endpoints via **Redis caching** and **response compression**, reducing average response latency by **10–15%** under production traffic.

Projects

GitHub Repos Manager — MCP Server | *Node.js, TypeScript, GitHub API, MCP*

[code](#)

- Built a lightweight **MCP server** that exposes GitHub repository operations to any MCP-compatible client without requiring heavy Docker dependencies or OAuth redirect flows.
- Implemented **89 GitHub API commands** out-of-the-box via environment-based token auth, enabling AI agents to automate repository workflows (PRs, issues, branches) programmatically and securely.
- Engineered robust rate-limiting and error-handling middlewares to strictly adhere to GitHub's API quotas, preventing request throttling during high-frequency parallel agent executions.
- Optimized payload serialization logic to strip redundant metadata, reducing network overhead by over 30% and accelerating the responsiveness of agent-to-server interactions.

Medical RAG: Retrieval-Augmented QA | *Python, Meditron-7B, LangChain, Qdrant, Ragas*

[code](#)

- Built a domain-specific **RAG system** for biomedical question answering over PubMed literature using 512-token chunks with 64-token overlap and **PubMedBERT** embeddings for high-precision retrieval.

- Orchestrated **Qdrant** vector retrieval (top-5) with **Meditron-7B** inference to generate context-grounded responses; evaluated pipeline with **Ragas** achieving **0.84+ faithfulness** score across test queries.
- Designed an asynchronous data ingestion pipeline to parse, clean, and embed thousands of complex medical PDFs and XML documents, significantly improving base vector quality.
- Implemented advanced retrieval techniques, including **Hybrid Search** (BM25 + Dense) and **Cross-Encoder Re-ranking**, to maximize context relevance for highly specialized clinical queries.

Multi-Agent RAG Customer Support | *Python, LangGraph, LangChain, Qdrant*

[code](#)

- Architected a **multi-agent conversational AI** with **LangGraph** for intent-based dynamic routing of complex travel queries to specialized domain-specific sub-agents with stateful memory across turns.
- Integrated **Qdrant** for fast vector retrieval and implemented conditional routing logic with full system observability via **LangSmith** tracing and token-level debugging.
- Developed a **Semantic Cache** using Redis to persist contextual dialog history and immediately resolve duplicate queries, cutting down redundant LLM API calls and reducing latency.
- Introduced a self-reflection and error-correction node within the graph loop, allowing agents to autonomously re-query the database or escalate to fallback logic upon detecting low-confidence generations.

Technical Skills

Languages: Python, Go, C++, TypeScript, SQL

AI & LLM Engineering: RAG, LangChain, LlamaIndex, LangGraph, NVIDIA Triton, vLLM, LLM Evaluation (**Ragas**)

Backend & Pipelines: FastAPI, Node.js, **Kafka**, Debezium, **gRPC**, RESTful API design, Microservices

Databases & Vector Stores: PostgreSQL, Cloud SQL, MongoDB, Redis, **Qdrant**, Milvus, Neo4j

Cloud & DevOps: GCP, Docker, Kubernetes, Git

Observability & Tooling: LangSmith, MCP servers, LangFuse

Achievements & Positions of Responsibility

- Rated **1721** at LeetCode with over **850 problems** solved, ranking among the top **10%** of Coders [Leetcode](#)
- Reached a peak rating of **1720 (4-star)** on CodeChef with a **global rank of 520** in the Starters 120 Competition.
- Secured **2nd Rank** in HackNITP 2.0 Hackathon, showcasing coding proficiency and critical thinking skills.
- Secured a perfect **10.0 CGPA** in the 8th semester, showcasing strong academic excellence.
- **Team Lead**, Google Developer Student Club – **Backend Team** **2023-2025**
- **Team Member**, Google Developer Student Club – **DSA Team** **2023-2025**